ClassAccess: Zoom-Based Real-Time Classroom Captioning

Team Name: Design for Social Devastation

1. MVP Definition and End-to-End Demonstration Plan

1.1 Project Overview

ClassAccess is a classroom accessibility system that converts speech from a live Zoom class into real-time captions displayed on a projector. It aims to help hearing-impaired students follow classroom instruction by providing readable text while the teacher conducts the class.

The Minimum Viable Product (MVP) for Hackathon 1 will focus exclusively on capturing Zoom audio, converting speech into text, and displaying captions on a projector with minimal delay. The prototype will demonstrate the complete pipeline rather than attempt to implement the full ClassAccess platform.

1.2 MVP Features

- **Real-Time Speech-to-Text:** Convert captured Zoom audio into text using a speech recognition model.
- **Live Caption Display:** Display recognized speech in large, readable text.
- **Automatic Caption Updates:** Update captions as new speech is recognized.
- **Full-Screen Projector Mode:** Provide a simple interface optimized for classroom projection.
- **Connection Status:** Indicate whether the captioning system is connected and receiving audio.

1.3 End-to-End Demonstration Plan

The demonstration will follow this workflow:

Zoom Class → Audio Capture → Speech-to-Text Model → Backend → WebSocket Communication → Caption Webpage → Classroom Projector

The demonstration will proceed as follows:

1. Start a Zoom meeting with a speaker presenting a short lesson.
2. Capture the meeting audio using system audio loopback or another suitable audio capture method.
3. Process the captured audio using the speech recognition model.
4. Send the recognized text through the backend to the caption webpage.
5. Display the captions in real time without refreshing the webpage.
6. Project the webpage and evaluate readability, transcription quality, and response time.

If live Zoom audio capture is not ready, a recorded test sample will be used to validate the transcription and caption display pipeline. The final objective remains a live, end-to-end demonstration.

1.4 Success Criteria

The MVP will be considered successful if:

- Audio from a Zoom session can be captured reliably.
- Spoken words are converted into readable text.
- Captions appear on the webpage automatically, without manual text entry.
- The caption webpage is readable when projected in a classroom.
- Transcription errors and caption latency can be evaluated using prepared test samples.
- The complete pipeline can be demonstrated and its limitations documented.

2. Implementation Plan and GitHub Mapping

2.1 Proposed System Architecture

The system consists of five main components:

1. **Audio Capture:** Captures audio from the Zoom meeting using system audio loopback, a virtual audio device, or a suitable direct integration.
2. **Speech Recognition:** Processes short audio chunks and converts speech into text using Faster-Whisper or another evaluated speech recognition model.
3. **Backend:** Uses Python and FastAPI to manage the transcription pipeline and deliver caption updates.
4. **Communication Layer:** Uses WebSockets to transmit recognized text to the frontend as it becomes available.
5. **Frontend and Projection:** Uses HTML, CSS, and JavaScript to display captions in a full-screen layout suitable for projection.

Direct Zoom integration through Real-Time Media Streams (RTMS) may be investigated if ordinary audio capture proves unsuitable. The initial implementation will prioritize the simplest reliable approach.

2.2 Implementation Tasks

Task 1: Build the Caption Display UI

Create a webpage that displays captions in large, readable text. Implement a full-screen layout, automatic caption updates, and a connection status indicator. Test readability from a distance.

Task 2: Set Up the FastAPI Backend

Create the Python FastAPI application and a WebSocket endpoint for caption delivery. Define a consistent message format and handle client connections and disconnections.

Task 3: Implement Speech-to-Text

Evaluate Faster-Whisper for local transcription. Process audio in short chunks, handle pauses and incomplete transcription segments, and evaluate transcription accuracy and latency.

Task 4: Capture Audio from Zoom

Test system audio loopback or a virtual audio device. Verify reliable audio capture and pass audio chunks to the transcription module. Investigate RTMS if direct Zoom integration becomes necessary.

Task 5: Prepare and Process Test Data

Generate synthetic speech or prepare short recordings with permission. Create reference transcripts, test different speaking speeds and mild background noise, and record transcription accuracy and latency.

Task 6: Integrate Live Captions End to End

Connect audio capture, speech recognition, the backend, WebSocket communication, and the caption webpage. Verify that recognized text reaches the frontend automatically and measure end-to-end latency.

Task 7: Test the Classroom Projector Setup

Run the system with a projector and evaluate caption readability, transcription errors, delays, and recovery from audio or network interruptions. Document known limitations and demonstration instructions.

The frontend and backend can be developed in parallel. Audio capture and speech recognition will be tested independently before integration. Audio capture is an early technical risk and will therefore be tested as soon as possible.

2.3 GitHub Issues and System Architecture Mapping

GitHub Issues will be used to track implementation tasks, assign responsibilities, and coordinate integration and testing. Each issue will include a description, a checklist of subtasks, and an assigned team member where appropriate.

GitHub issue	Main tasks	Corresponding architecture component
Issue 1: Build Caption Display UI	Caption layout, typography, full-screen mode, connection status	Frontend and projector display
Issue 2: Set Up the FastAPI Backend	Backend setup, WebSocket endpoint, message format, connection handling	Backend and communication layer
Issue 3: Implement Speech-to-Text	Model evaluation, chunk processing, transcription and latency testing	Speech recognition

Issue 4: Capture Audio from Zoom	Audio loopback, audio capture, interruption handling, RTMS investigation	Zoom audio capture
Issue 5: Prepare and Process Test Data	Synthetic speech, reference transcripts, test conditions, evaluation	Data preparation and testing
Issue 6: Integrate Live Captions End to End	Connect audio capture, transcription, backend, and frontend	End-to-end system integration
Issue 7: Test the Classroom Projector Setup	Projector testing, readability, latency, error handling	Complete system validation

The issues will serve as the execution plan for development. Progress will be tracked through GitHub, and the implementation will follow the dependencies between components rather than treating every task as independent.

3. (Synthetic) Data Follow-up

3.1 Data Plan and Purpose

The test data will be used to verify that the speech recognition model can convert classroom-style speech into text and that the captioning interface can display recognized text with minimal delay.

A small dataset will be sufficient for preliminary testing of the Hackathon 1 prototype. The primary data source will be synthetically generated speech, supplemented by short recordings made with permission if required.

The data plan will cover four conditions: clear speech, different speaking speeds, natural pauses, and mild background noise.

3.2 Data Generation

The team will prepare short educational scripts containing ordinary classroom vocabulary and complete sentences. These scripts will be converted into audio using a text-to-speech tool.

The generated samples will include:

- **Clear speech:** Normal-speed speech for baseline transcription testing.
- **Different speaking speeds:** Slower and faster delivery to examine changes in recognition quality.
- **Natural pauses:** Pauses between sentences to test caption updates and sentence boundaries.

- **Mild background noise:** Samples with controlled background noise to identify transcription errors under less ideal conditions.

If synthetic speech does not adequately represent the intended test conditions, a small number of human speech recordings may be collected with the speakers' permission.

3.3 Data Procurement

The project does not initially require purchasing a dataset or collecting a large external speech corpus.

The planned sources are:

1. **Primary source:** Synthetic speech generated from team-prepared educational scripts.
2. **Supplementary source:** Short speech recordings produced by consenting volunteers, if necessary.
3. **Live validation source:** Audio from a test Zoom meeting conducted specifically for the demonstration.

If an external audio sample is required, the team will use only material with appropriate reuse permissions and document its source and licence. External data will not be necessary if the synthetic samples and test meeting provide sufficient coverage.

3.4 Data Processing and Evaluation

The data processing workflow will be:

1. Prepare short educational scripts.
2. Generate synthetic audio or record authorized speech samples.
3. Convert the audio into a format supported by the transcription pipeline.
4. Maintain a reference transcript for each sample.
5. Run each sample through the speech recognition model.
6. Compare the recognized text against the reference transcript.
7. Measure end-to-end latency between speech input and caption display.
8. Record errors and compare results across the different test conditions.

Evaluation metrics:

- **Word Error Rate (WER):** Measures transcription errors relative to the reference transcript.
- **End-to-end latency:** Measures the delay between spoken input and the corresponding displayed caption.
- **Display reliability:** Checks whether captions appear and update correctly without refreshing the webpage.

The results will help identify whether transcription quality, audio capture, or caption delivery requires further improvement.

3.5 Data Quantum: Required Data Volume for Hackathon 1

The initial target is approximately **8–10 minutes of test audio**, with an additional five-minute live Zoom demonstration.

Data category	Planned quantity	Purpose
Clear synthetic speech	3 minutes	Establish baseline transcription performance
Speech at different speeds and with pauses	3 minutes	Test variations in speaking patterns
Speech with mild background noise	2–4 minutes	Identify transcription errors under noisy conditions
Live Zoom demonstration	5 minutes, separately	Validate the complete end-to-end pipeline

The first three categories account for approximately 8–10 minutes of test audio. The live Zoom demonstration is additional and will validate the integrated system rather than expand the prepared dataset.

This quantity is intended for preliminary functional testing, not for training a new speech recognition model or claiming comprehensive performance across all classroom environments. The dataset may be expanded if initial testing reveals important gaps.

3.6 Privacy and Data Handling

The project will follow basic privacy and data-minimization practices:

- Use synthetic speech wherever possible.
- Obtain permission before recording volunteers or capturing meeting audio for testing.
- Do not use identifiable classroom recordings involving minors without appropriate consent and authorization.
- Avoid collecting personal, sensitive, or unnecessary information.
- Restrict access to recordings and transcripts to the project team.
- Use test-specific recordings rather than capturing unrelated meetings.
- Delete temporary audio files when they are no longer needed.
- Do not upload private recordings or identifiable transcripts to a public GitHub repository.
- Store only the data required to evaluate transcription accuracy and caption latency.

These measures will reduce privacy risks while allowing the team to test the MVP under controlled conditions.

4. Future Scope

The following features are outside the Hackathon 1 MVP and may be considered in future iterations:

- Improve transcription accuracy in noisy classroom environments.
- Reduce the delay between speech and caption display.
- Add speaker identification to distinguish between teachers and students.
- Support regional languages and multilingual classrooms.
- Generate class transcripts, summaries, and revision notes.
- Introduce searchable transcripts and access to previous classes.
- Explore more reliable direct Zoom integration.
- Conduct classroom trials and gather feedback from hearing-impaired students and teachers.

5. Expected Outcome

The expected outcome is a working prototype that captures speech from a live Zoom class, converts it into text, and displays captions on a classroom projector with minimal delay.

The project will establish a GitHub-based workflow for implementation, data preparation, task allocation, integration, and testing. The prepared dataset will support preliminary evaluation of transcription accuracy and latency, while the live demonstration will validate the complete pipeline.

The MVP will demonstrate the feasibility of real-time speech recognition for classroom accessibility and identify the technical and privacy considerations that must be addressed before classroom deployment.